DATE:07/02/2026

EX-03

Network Security Essentials

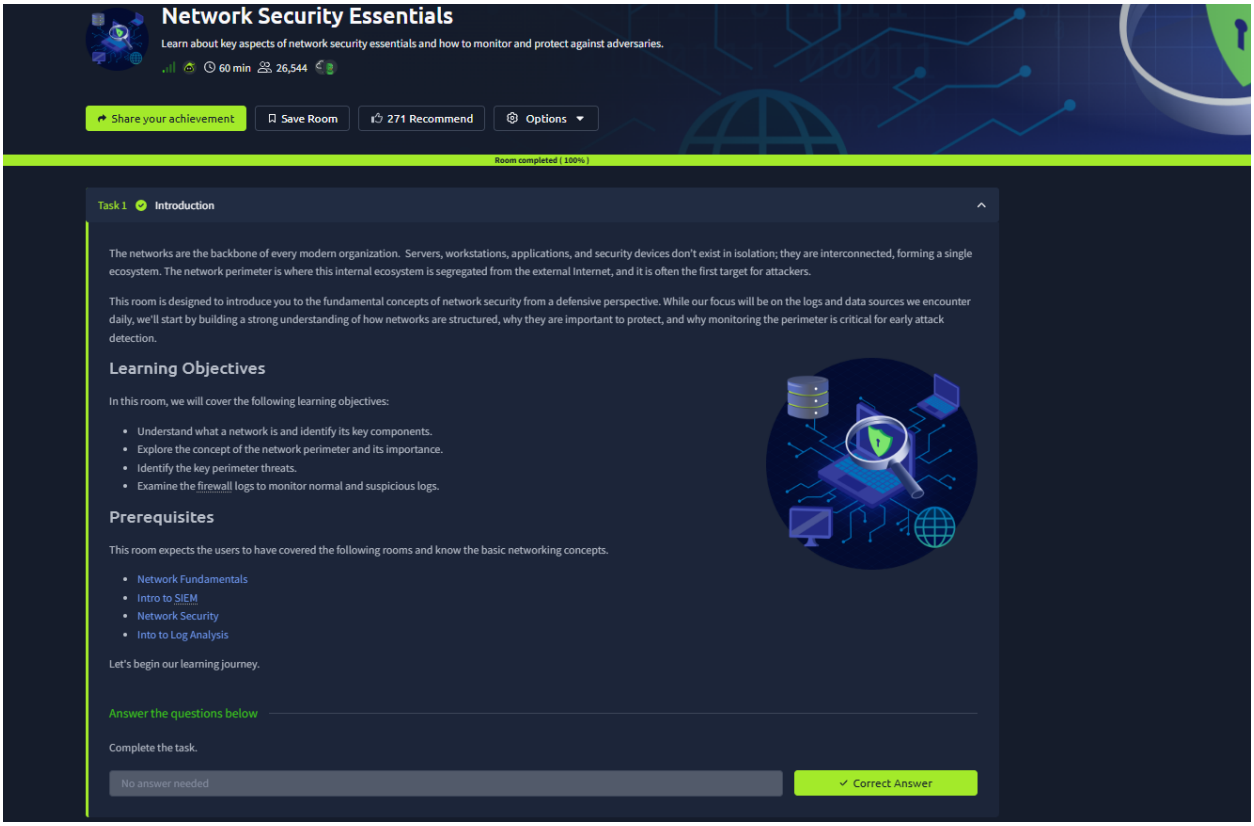
AIM:

To Understand network structure , visibility, perimeter security, monitoring, and log-based breach investigation.

PROCEDURE:

- Connected to the lab environment.
- Studied network overview and visibility.
- Learned network perimeter protection and monitoring.
- Analysed perimeter logs to investigate a security breach.

TASK-1:



TASK-2:

Task 2 Lab Connection

Before moving forward, start the lab by clicking the `Start Machine` button. It will take 3 minutes to load properly. The VM will be accessible on the right side of the split screen. If the VM is not visible, use the blue Show Split View button at the top of the page.

Set up your virtual environment

To successfully complete this room, you'll need to set up your virtual environment. This involves starting the Target Machine, ensuring you're equipped with the necessary tools and access to tackle the challenges ahead.

Target machine

Status: Off

Start Machine

All files required to complete the room are placed in the `Perimeter_logs` folder on the Desktop. The log files to be used in the task 7 for the investigation are pre-ingested into Splunk Instance, which can be accessed from within the Machine at `MACHINE_IP:8000`.

Answer the questions below

Connect to the Lab.

No answer needed

Correct Answer

TASK-3:

- Multiple failed login attempts (password spraying).
- Unusual logins from external IPs or at odd hours.
- Accounts accessing systems they normally shouldn't.

Routers & Switches (Network Infrastructure)

Routers connect different networks, most importantly linking the enterprise LAN to the Internet. Switches connect devices within the same network, ensuring employees' PCs, printers, and servers communicate seamlessly. These devices are the circulatory system of the enterprise.

Importance

While routers and switches are not directly exposed in most enterprise setups, if compromised, they provide attackers with the ability to:

- Intercept and manipulate traffic (Man-in-the-Middle attacks).
- Create backdoors by rerouting traffic.
- Open hidden channels to the Internet.

Firewalls / Perimeter Devices

A firewall is the primary security gateway that controls traffic between the trusted internal network and the untrusted Internet. It inspects incoming and outgoing packets and decides whether to allow or block them based on defined security rules. Modern firewalls also perform deep inspection of applications, intrusion prevention, and even malware detection.

Importance:

- Protects the enterprise from direct exposure to the Internet.
- Prevents unauthorized access to internal services (e.g., database or RDP).
- Logs every connection attempt, successful or blocked. These logs are often the earliest indicators of attacks such as port scans, brute-force attempts, or exploitation attempts.

Now that we understand what a network is, what components connect to form it, and their usage and importance, let's move to the next task and explore the key network perimeters.

Answer the questions below

Continue to the Next task.

No answer needed

Correct Answer

TASK-4:

- **What they show:** Source and destination IP, ports, protocols, and the action taken (e.g., allowed or blocked).
- **Why they are important:** They give you the crucial "when," "where," and "how." They can reveal an attacker's initial reconnaissance attempts, lateral movement between systems, or attempts to exfiltrate data.

To get a complete picture, you need to correlate both. Think of it this way: Host-centric logs tell you what happened inside a room, while network-centric logs tell you who entered and left the building.

Key Network-Centric Log Sources

- **Firewalls:** The firewall logs provide a record of every connection allowed or denied based on pre-defined security rules. Firewall logs are the first place to look for unauthorized connection attempts from the internet.
- **Intrusion Detection/Prevention Systems (IDS/IPS):** They monitor network traffic for patterns that match known malicious attacks (signatures) or for unusual behavior (anomalies). Their logs are critical for detecting active attacks in real time.
- **Routers and Switches:** While not logging in the traditional sense, devices like routers can generate flow data. This data summarizes traffic conversations, telling who talked to whom, for how long, and how much data was sent. It's excellent for getting a high-level overview of network activity.
- **Web Proxies:** These logs are a goldmine for organizations that route their web traffic through proxies. They record every website user's visit, providing visibility into web-based threats, policy violations, or data exfiltration attempts.
- **VPN:** These devices manage remote access for employees. Their logs track who is connecting to the corporate network, from where, and at what time, which is essential for monitoring the security of remote connections.

▶ **Importance of Network-Centric Logs (Expand to read)**

Effective security monitoring relies on examining both host and network logs. This combined approach empowers analysts to assemble a thorough and accurate timeline of any security incident.

In this task, we have explored the network visibility, which is the key concept of network security. In the next task, we will explore network perimeters and how they help us monitor and protect the network.

Answer the questions below

Continue to the next task.

No answer needed

✓ Correct Answer

TASK-5:

- Employees outside the office connect through a VPN gateway.

This setup ensures that only controlled traffic reaches the internal network, while public services remain isolated in a safer zone.

Commonly includes:

- Routers: Direct network traffic.
- Firewalls: Inspect and filter traffic.
- DMZ (Demilitarized Zone): Hosts public-facing services.
- VPN Gateways: Secure remote access.

Why does it matter

The perimeter is the first line of defense against external threats. Attackers scan perimeter IPs looking for open ports and vulnerabilities. Misconfigured or weakly protected perimeters often lead to:

- Unauthorized access (e.g., exposed RDP/SSH)
- Data breaches
- Malware infiltration

Role of a Security Analyst

As a security analyst, monitoring the perimeter means:

- Reviewing firewall logs for blocked/allowed connections.
- Identifying scanning attempts or brute-force login attempts.
- Flagging unusual outbound traffic that may indicate malware beaconing or exfiltration.
- Understanding what should (and should not) be exposed at the perimeter.

Answer the questions below

Continue to the next task.

No answer needed

✓ Correct Answer

TASK-6:

- To find the attack, an analyst would filter or group the logs by source IP.
- Doing so would immediately reveal that a suspicious user is responsible for hundreds of failed login attempts in a very short period.
- Analyst's Verdict: One suspicious IP is observed conducting a brute-force attack against the VPN gateway. The attacker is attempting to use a list of common usernames ('admin', 'root', 'test', etc.) to find a valid account to compromise. The scattered successful logins are benign traffic from legitimate employees.

Key Takeaways

- An analyst's job is to separate normal traffic from suspicious activity.
- Look for suspicious patterns.
- Repetition from one source to many destinations = Scanning.
- Repetition from one source to one destination = Brute-forcing.
- Traffic at perfect, regular intervals = Malware beaconing.
- Context is key. An IDS alert telling you why something was flagged is more valuable than a simple firewall block.
- Monitoring the network perimeter is the first step in detecting attacks.

Answer the questions below

Examine the firewall logs. Which IP address is performing the port scan?

203.0.113.10 — ✓ Correct Answer

In the WAF Logs, which single source IP is responsible for all the blocked web attacks?

198.51.100.12 — ✓ Correct Answer

In the VPN logs, how many brute-force attempts failed?

90 — ✓ Correct Answer

Which suspicious IP address was found attempting the brute-force attack against the VPN gateway?

45.137.22.13 — ✓ Correct Answer

TASK-7:

Examine the firewall logs. What external IP performed the most reconnaissance?

203.0.113.45 — ✓ Correct Answer

In the firewall log, Which internal host was targeted by scans?

10.0.0.20 — ✓ Correct Answer

Which username was targeted in VPN logs?

svc_backup — ✓ Correct Answer

What internal IP was assigned after successful VPN login?

10.8.0.23 — ✓ Correct Answer

Which port was used for lateral SMB attempts?

445 — ✓ Correct Answer

In the IDS logs, which host beaconed to the C2?

10.0.0.60 — ✓ Correct Answer

During the investigation, which IP was observed to be associated with C2?

198.51.100.77 — ✓ Correct Answer

Which host showed the exfiltration attempts?

10.0.0.51 — ✓ Correct Answer

TASK-8:

Task 8 Conclusion

In this room, you learned that an enterprise network is not just a collection of computers and servers. It's an ecosystem with critical components like firewalls, Active Directory, application servers, file/database servers, endpoints, and wireless access points.

- The network perimeter acts as the boundary between trusted and untrusted networks.
- Attackers test this boundary constantly with port scans, brute-force attempts, and exploits against exposed services.
- Monitoring the perimeter allows SOC analysts to detect these early signs before attackers can move deeper into the network.

As a security analyst, our role is to:

- Recognize normal vs suspicious network traffic.
- Escalate unusual activity to higher SOC tiers.
- Understand how each network component fits into the bigger picture of enterprise defense.

Answer the questions below

Complete the room.

No answer needed

✓ Correct Answer

RESULT:

Successfully Learned basic network Security Conepts, Perimeter defense and log analysis through hands on practice.